

CHANGDA, Russian Federation

WMO: 247390

Lat: 62.283N

Lon: 119.850E

Elev: 1011

StdP: 14.17

Time Zone: 9.00 (E09)

Period: 83-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -58.3	(c) -53.9	(d) -61.3	(e) 0.1	(f) -58.0	(g) -57.1	(h) 0.2	(i) -53.7	(j) 9.4	(k) -10.5	(l) 7.7	(m) -12.9	(n) 0.2	(o) 180	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.6	84.2	64.5	79.5	62.5	75.1	59.9	67.1	80.8	64.4	76.2	61.8	72.0	3.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.9	85.8	72.0	59.6	79.0	69.2	57.4	72.9	67.2	32.3	82.2	30.1	76.8	28.2	72.3	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 11.8	(b) 9.8	(c) 7.9	(e) -62.7	(f) 89.0	(g) 4.9	(h) 3.1	(i) -66.2	(j) 91.3	(k) -69.0	(l) 93.1	(m) -71.8	(n) 94.9	(o) -75.3	(p) 97.1		
			DB			3.7	-66.2	72.5	-69.0	74.6	-71.8	76.7	-75.3	79.4		
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	-29.5	-18.9	0.5	20.2	41.4	55.5	61.0	54.9	41.0	18.5	-10.5	-24.3	
	DBStd	34.23	15.39	14.87	14.60	12.06	9.59	7.71	7.96	6.72	8.00	11.95	15.66	17.23	
	HDD50	12573	2465	1930	1535	894	301	29	7	27	289	977	1816	2303	
	HDD65	17350	2930	2350	2000	1344	733	299	177	321	720	1442	2266	2768	
	CDD50	771	0	0	0	0	33	194	348	178	18	0	0	0	
	CDD65	75	0	0	0	0	0	14	54	7	0	0	0	0	
	CDH74	1201	0	0	0	0	13	303	764	120	1	0	0	0	
	CDH80	357	0	0	0	0	2	66	262	27	0	0	0	0	
Wind	WSAvg	2.8	1.6	2.3	3.2	3.6	3.6	3.3	2.9	2.7	2.9	3.0	2.5	2.3	
Precipitation	PrecAvg	15.2	0.7	0.6	0.5	0.6	1.4	1.9	2.6	2.2	1.6	1.2	1.1	0.8	
	PrecMax	21.1	1.2	1.2	0.8	1.4	2.5	3.9	6.1	3.7	3.3	2.2	1.8	1.6	
	PrecMin	10.6	0.3	0.2	0.1	0.3	0.5	1.0	0.9	0.8	0.7	0.4	0.4	0.2	
	PrecStd	2.3	0.2	0.2	0.2	0.3	0.6	0.6	1.3	0.8	0.6	0.4	0.4	0.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	19.7	41.2	53.6	75.7	85.8	89.8	84.2	73.0	47.7	32.9	21.9	
		MCWB	10.6	17.0	34.8	40.4	55.7	63.0	66.9	67.6	56.9	39.0	30.4	21.0	
	2%	DB	5.0	14.1	34.6	47.4	70.0	81.1	86.9	76.7	65.0	42.0	25.2	16.3	
		MCWB	4.4	13.0	29.2	36.6	51.6	60.5	66.7	61.7	52.5	37.3	24.3	15.6	
	5%	DB	0.1	8.6	29.3	43.8	64.8	77.0	82.4	73.3	58.8	37.5	20.3	9.7	
		MCWB	-0.5	7.7	26.0	34.5	48.8	58.5	65.2	59.4	48.9	34.1	19.6	9.0	
	10%	DB	-5.9	3.0	23.9	39.8	59.2	72.8	77.4	69.4	54.0	34.3	14.1	2.3	
		MCWB	-6.4	2.3	21.2	32.3	46.3	57.1	62.6	57.9	46.0	31.7	13.5	1.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	17.4	35.6	40.9	57.7	65.0	70.7	67.3	57.7	40.4	30.8	19.5
			MCDB	10.9	19.8	40.2	51.3	74.6	81.4	83.6	82.7	71.0	45.2	32.3	20.1
2%		WB	4.5	13.0	29.7	37.5	53.0	62.5	68.5	63.7	53.8	37.9	24.3	15.2	
		MCDB	4.9	14.3	34.0	46.2	67.1	76.7	82.3	73.4	63.6	42.0	25.1	15.9	
5%		WB	-0.4	7.7	25.9	35.3	49.9	60.3	66.7	61.4	50.4	34.6	19.5	8.7	
		MCDB	0.2	8.6	29.2	42.6	61.8	73.5	79.8	70.2	57.4	37.1	20.3	9.4	
10%		WB	-6.4	2.3	21.3	32.7	47.0	58.0	64.2	59.2	47.1	32.0	13.4	1.4	
		MCDB	-6.0	3.0	23.7	39.2	58.8	70.3	75.5	67.2	52.9	34.2	13.9	2.0	
Mean Daily Temperature Range	5% DB	MDBR	16.1	21.3	26.1	24.6	21.0	24.2	23.6	23.6	18.3	14.7	18.2	15.9	
		MCDBR	18.9	21.9	21.7	26.9	29.6	32.4	29.7	29.7	26.3	13.8	19.9	17.9	
		MCWBR	18.4	21.3	17.7	17.9	15.6	14.2	12.6	15.3	15.0	9.5	20.5	17.9	
	5% WB	MCDBR	18.0	21.7	20.8	24.3	25.9	27.8	26.8	24.9	25.0	13.1	20.5	17.8	
		MCWBR	17.6	20.9	17.3	16.4	15.2	13.8	12.6	13.9	14.7	10.0	20.5	17.6	
Clear-Sky Solar Irradiance	taub	0.220	0.259	0.290	0.331	0.371	0.404	0.488	0.439	0.327	0.277	0.215	0.190		
	taud	1.942	2.062	2.095	2.125	2.276	2.292	2.292	2.017	2.125	2.455	2.337	1.980	1.873	
	Ebn at Noon	137	218	256	269	268	259	229	230	248	216	137	85		
	Edh at Noon	12	23	32	39	37	37	48	39	23	17	11	7		
All-Sky Solar Radiation	RadAvg	105	414	970	1589	1687	1807	1649	1253	780	358	148	50		
	RadStd	7	21	36	73	107	118	101	77	88	32	16	2		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	+1.30	N/A	N/A	N/A	N/A	N/A	-409	-477	N/A	N/A

Nomenclature: See separate page